



P. E. S. COLLEGE OF ENGINEERING
MANDYA – 571401

(An Autonomous Institution Affiliated to VTU, Belgaum)



MINI PROJECT REPORT

ON

“UrbanPulseAI – Smart Traffic Intelligence and Decision Support System”

SUBMITTED IN PARTIAL FULFILMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING

IN

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

SUBMITTED BY

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2025-2026



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DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

CERTIFICATE

This is to certify that Mr. Ashutosh Raj [4PS23CB005] student of VI semester B.E, **DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS**, P.E.S. College of Engineering Mandya have satisfactorily completed the **MINI PROJECT (P22CBMP607) ON “UrbanPulseAI – Smart Traffic Intelligence and Decision Support System”** during the academic year 2025-26. The Project report has been approved as it satisfies the academic requirements of VI semester B.E Computer Science and Business Systems discipline.

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DECLARATION

I **Ashutosh Raj [4PS23CB005]** student of 6th semester pursuing **BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND BUSINESS SYSTEMS (CS&BS)**, hereby declare that the project report titled "**UrbanPulseAI – Smart Traffic Intelligence and Decision Support System**" is an original work carried out by us during the conduction period. We affirm that this report has not been submitted to any other university or institution for the award of any degree, diploma, or certificate. The information presented in this report is authentic based on our personal experience & learning and submitted in partial fulfilment of the requirements for the award of degree in B.E. Computer Science and Business Systems affiliated to Visvesvaraya Technological University (VTU) Belagavi during the academic year 2025-2026.

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TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
1	INTRODUCTION	1-4
	1.1 BACKGROUND	1-3
	1.2 PROBLEM DEFINITION	3-4
	1.3 OBJECTIVES	4
	1.4 CHAPTER ORGANIZATION	4
2	LITERATURE REVIEW	5-13
	2.1 EXISTING WORKS	5-8
	2.2 TECHNIQUES USED	8-11
	2.3 IDENTIFIED GAPS	11-13
3	METHODOLOGY AND IMPLEMENTATION	14-30
	3.1 TOOLS AND TECHNOLOGIES	14-16
	3.2 SYSTEM REQUIREMENTS	17
	3.3 FUNCTIONAL REQUIREMENTS	17
	3.4 NON FUNCTIONAL REQUIREMENTS	17
	3.5 SYSTEM ARCHITECTURE	17-22
	3.6 MODULES DESCRIPTION	23-25
	3.7 SCREENSHOTS	25-28
	3.8 EVALUATION & TESTING	29-31
4	RESULTS AND DISCUSSION	32-38
	4.1 OUTPUTS	32-34
	4.2 PERFORMANCE ANALYSIS	34-35
	4.3 COMPARISON	35-38
	4.4 INTERPRETATION	38-39
5	CONCLUSION	40-42
	5.1 LIMITATIONS	42
	5.2 FUTURE SCOPE	42
6	REFERENCES	43-44

LIST OF FIGURES

FIG NO.	CAPTION	PAGE NO
3.1	System Architecture Of Urbanpulse AI	22
3.2	Citizen Portal For Incident Reporting	25
3.3	AI Based Incident Verifications	26
3.4	Authority Dashboard Interface	27
3.5	Traffic Prediction And Congestion Visualization	27
3.6	Optimized And Emergency Routing	28
3.7	AI Copilot Interface	28
4.1	AI Driven Consequence Analysis And Network Impact	32
4.2	Performance Analysis Of UrbanPulse AI	35

LIST OF TABLES

TABLE NO.	CAPTION	PAGE NO
3.1	Hardware Requirements Of UrbanPulse AI	17
3.2	Software Requirements Of UrbanPulse AI	17
3.3	Functional Validation of UrbanPulse AI	31
4.1	Summary Of Consequence Analysis Metrics	33
4.2	Comparison Of UrbanPulse AI With Reference Studies	37
4.3	Comparison Of UrbanPulse AI With Reference Applications	38

ABSTRACT

Traffic congestion has become one of the most common problems faced by people in modern cities. Increasing numbers of vehicles, rapid urban growth, road accidents, flooding, potholes, and other unexpected incidents often cause delays, inconvenience, and safety concerns for commuters. Traffic authorities also face challenges in identifying problems quickly and responding effectively, especially when information is scattered across multiple sources. As cities continue to grow, there is a need for smarter solutions that can help both citizens and authorities manage transportation more efficiently.

UrbanPulseAI is a smart city platform designed to improve traffic management and urban mobility. The system continuously monitors traffic conditions and helps identify areas where congestion is likely to occur. Citizens can actively participate by reporting incidents such as accidents, road damage, flooding, or road blockages through a simple reporting interface. To improve the reliability of these reports, uploaded images are automatically checked before the complaint is forwarded to the concerned authorities. The platform also helps users find better travel routes, provides traffic updates, and supports authorities with useful information for handling incidents and managing transportation services more effectively.

By bringing together traffic monitoring, citizen participation, incident management, and decision-support capabilities within a single platform, UrbanPulseAI aims to create a safer, smoother, and more efficient transportation environment. The system encourages collaboration between citizens and authorities, enables faster response to road-related problems, and supports better planning and management of urban transportation. Ultimately, it contributes towards building smarter and more sustainable cities while improving the daily travel experience of the public.